

*Cycle journey planning,
for cyclists, by cyclists*



Is the OSM data model creaking?

Slides at: www.cycliststreets.net/blog

PROBLEM: Compromises from
OSM representing spaces as lines.





One conceptual space, but multiple flows

	Foot Path	i
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[Zoom to this](#)

▼ All fields

Name	i
Common name (if any)	+

Surface		i
asphalt		

Width (Metres)		i
5		

Structure		i
☐ Bridge		



Two tunnels, as two lines, one unnamed



How do we model this?

▼ All tags (2)



highway	▼	primary	▼		
cycleway	▼	track	▼		
+					



0. Original method c. 2008

▼ All tags (12)



cycleway	▼	track	▼		
foot	▼	no	▼		
highway	▼	cycleway	▼		
lcn	▼	yes	▼		
lit	▼	yes	▼		
name	▼	Hills Road (cycle...	▼		
oneway	▼	yes	▼		
segregated	▼	light	▼		
sidewalk	▼	left	▼		
surface	▼	asphalt	▼		
surface:colour	▼	red	▼		
width	▼		▼		



1. Separate paths approach

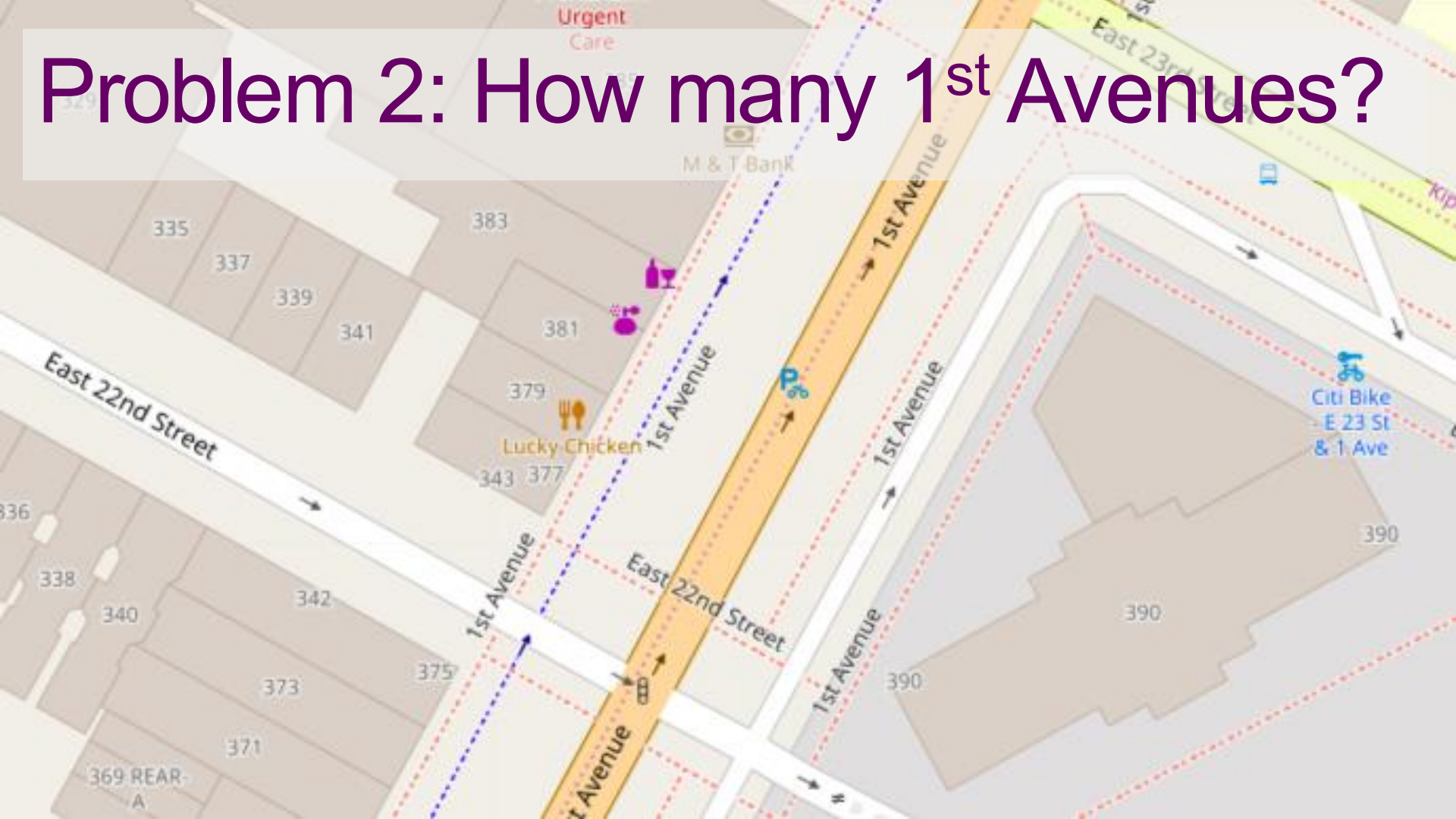


In reality you
can cross the
road.

No-one
actually
cycles like
this:



Problem 2: How many 1st Avenues?



2. Unified street approach

access:lanes=no|no|no|yes|yes|no|no

bicycle:lanes=no|designated|designated|yes|yes|designated|no

bus:lanes=no|no|no|yes|yes|no|no

cycleway:backward=track

cycleway:backward:est_width=1.5

cycleway:backward:oneway=-1

cycleway:backward:segregated=no

cycleway:left=stepped

cycleway:left:oneway=yes

cycleway:left:width=2.1

cycleway:right=stepped

cycleway:right:oneway=yes

cycleway:right:width=2.1

est_width=6

foot:lanes=yes|no|no|no|no|no|yes

highway=primary

lanes=2

lanes:backward=1

lanes:bicycle=3

lanes:foot=2

lanes:forward=1

lcu=no|no|yes|no|no|yes|no

lit=yes

maxspeed=30 mph

name=Hills Road

note=there are cycle lanes in both directions PLUS a separate cycle track

ref=A1307

source:lit=2011-03-12

surface=asphalt

Surface:color=black|red|black|black|red|black

taxi:lanes=no|no|no|yes|yes|no|no

An aerial photograph of a city street intersection. A white road with black arrows runs diagonally from the top-left to the bottom-right. A yellow road with black arrows runs diagonally from the top-right to the bottom-left. They intersect at a point marked with a small grey circle. A blue and white dashed line runs parallel to the yellow road. A street sign with the number '8' is visible on the white road. Text labels '1st Avenue' and 'East 13th Street' are visible on the roads. A semi-transparent grey box with purple text is overlaid on the right side of the image.

Where should
good walk
routing put the
pedestrian?
Sidewalk next to
road is wrong.

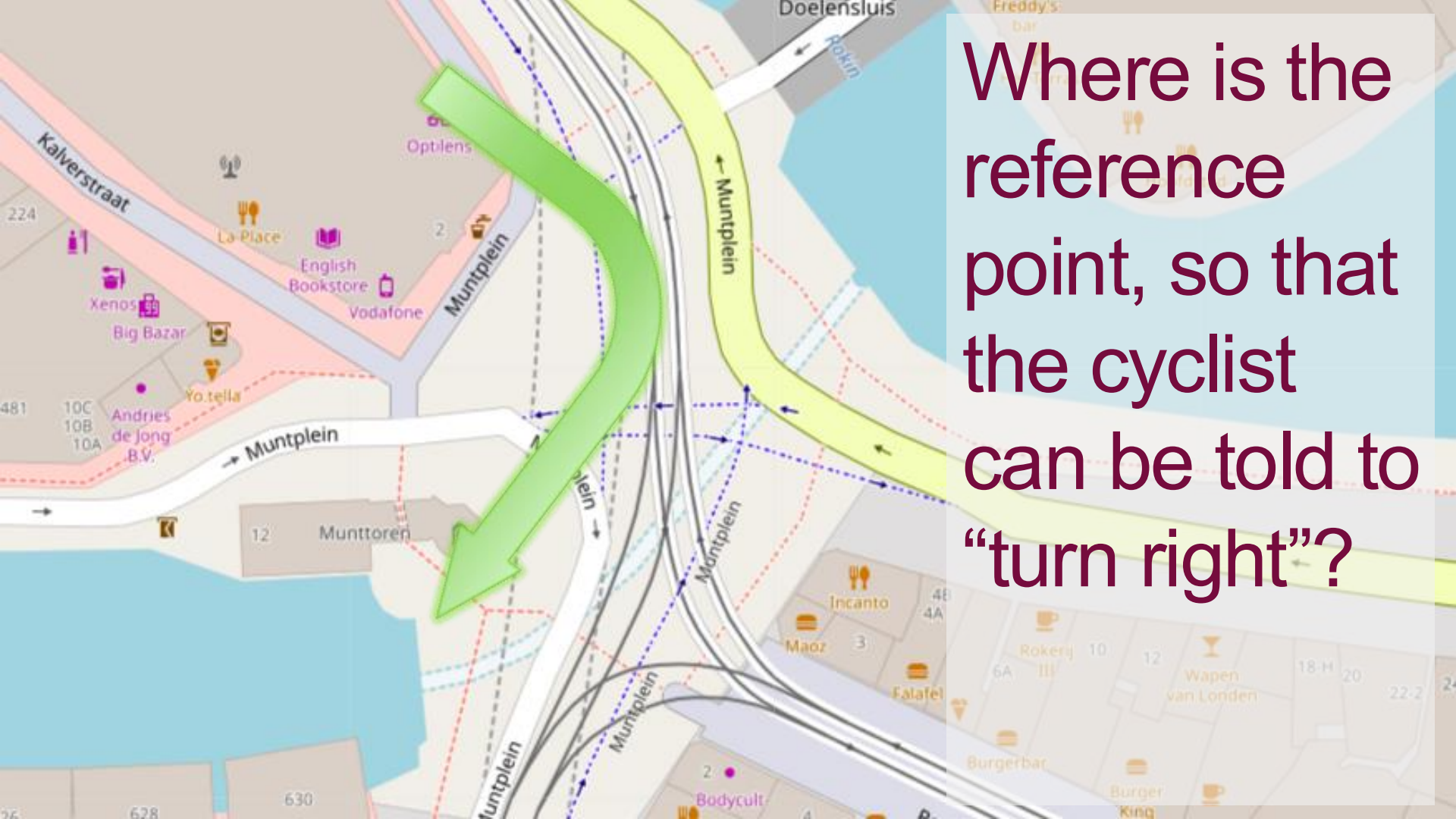
Problem: Can't properly model turns





We don't
really have
a semantic
concept of
“junction”,
just lines &
connections

Where is the reference point, so that the cyclist can be told to “turn right”?



Problem: Junction times unmeasurable

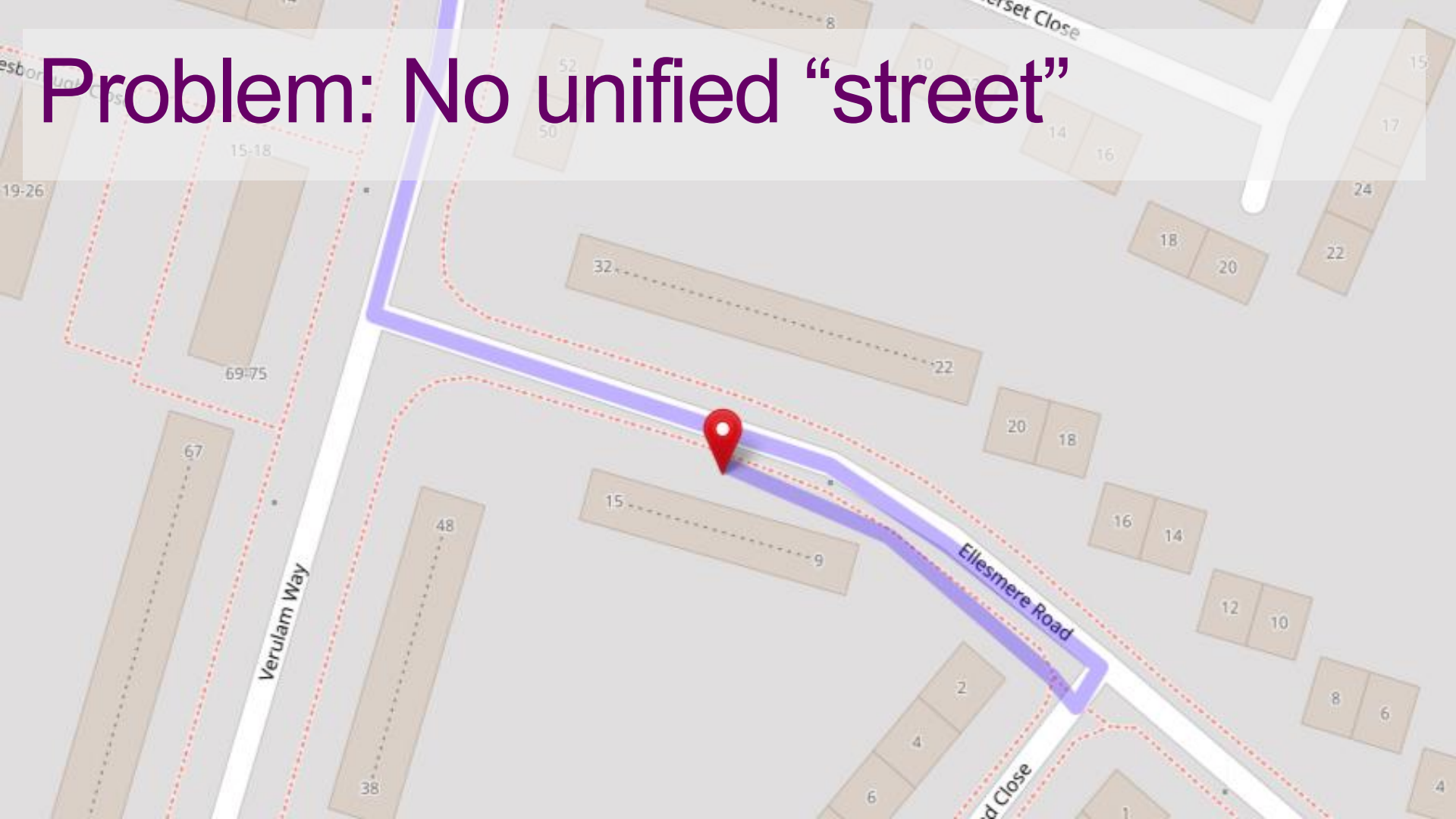


How many traffic lights for the cyclist turning right?



Is this what
humans really
perceive?

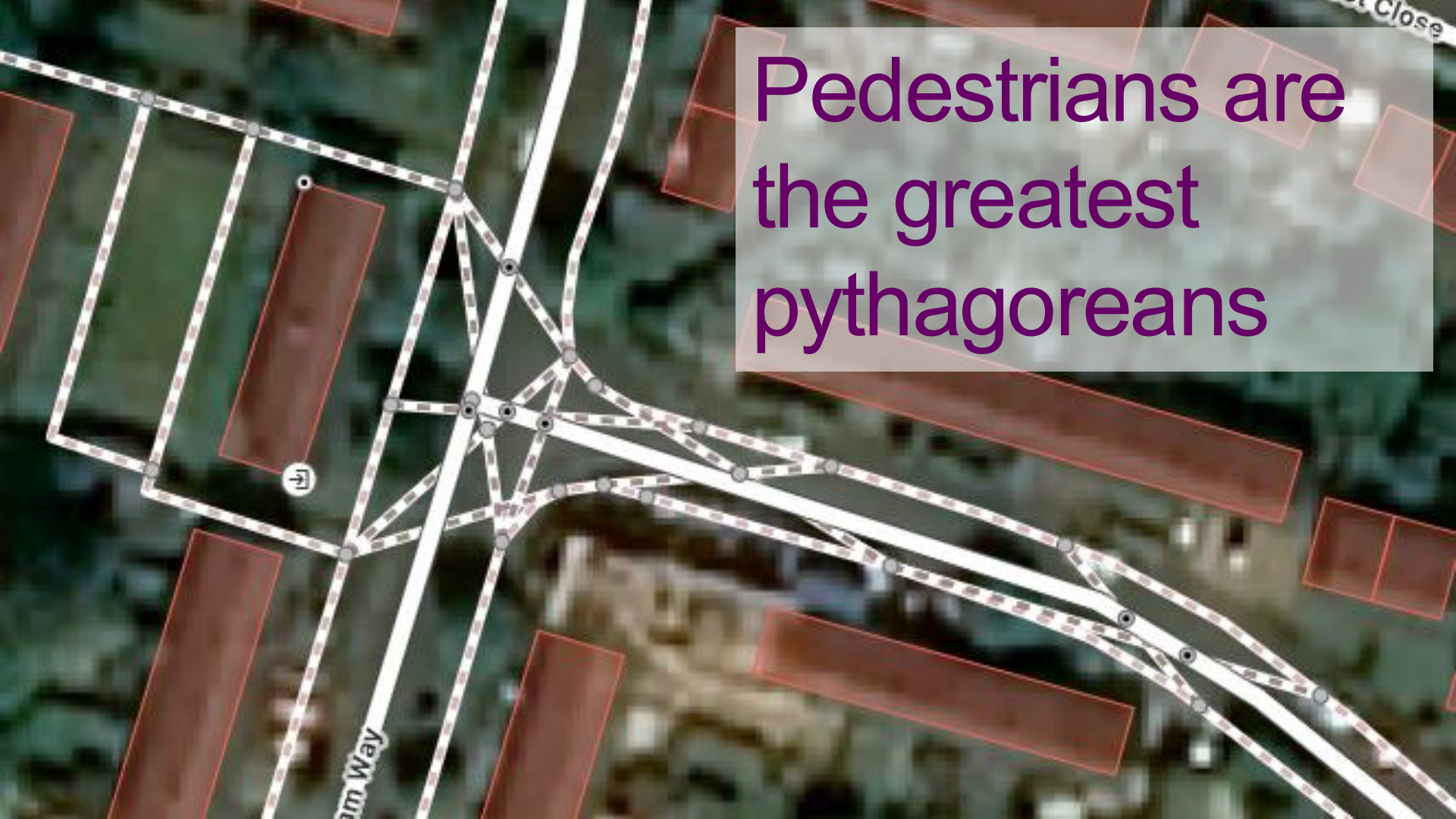




Problem: No unified “street”

The background is a map with a road network. A solid white line represents a main road, with a dashed white line indicating a specific route. The route starts at a junction on the left, goes up, then right, then down, and finally right again. It features several turns and a series of small, dark, circular markers along its path. A small white circle with a black 'E' is located near the start of the route. Several brown rectangular shapes are overlaid on the map, possibly representing buildings or other features. A semi-transparent grey box with purple text is in the upper right corner.

If routable, then
shouldn't it be
this?

An aerial photograph of a suburban area with houses and trees. A network of white lines with red dashed borders is overlaid on the map, connecting various points. Several red rectangles are placed over different areas of the map, possibly representing specific locations or zones. The text 'Pedestrians are the greatest pythagoreans' is overlaid on the right side of the image in a purple font.

Pedestrians are
the greatest
pythagoreans



Concept of
“Junction”

Surrounds all
relevant
features,
unifying them

▼ Feature Type

	Junction	i
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▼ Fields

Name	i
Common name (if any)	+

Add field:

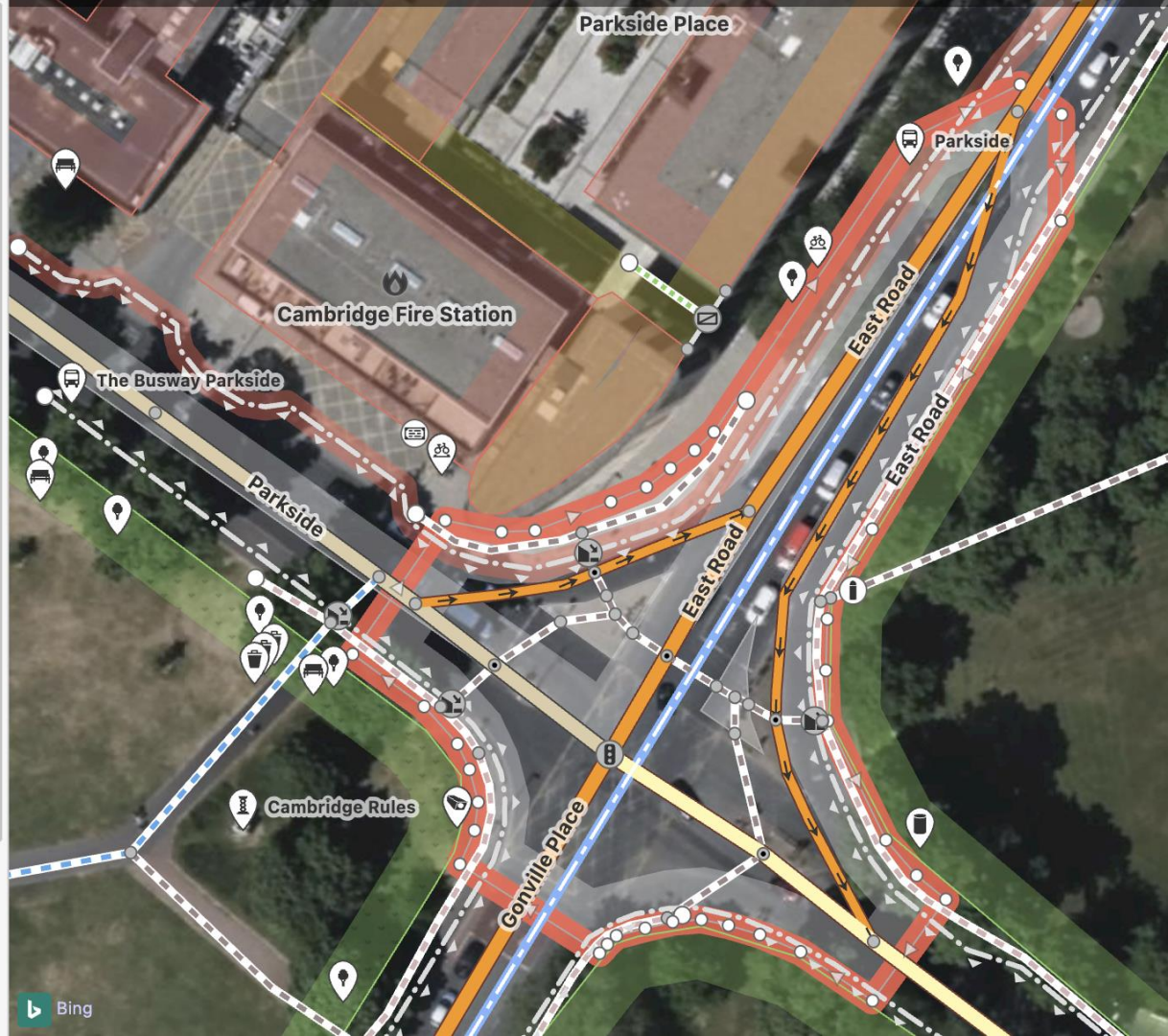
▼ Tags (2)



area=yes
junction=yes

▼ Relations (0)

East Road, Cambridge
www.openstreetmap.org/way/1069011490



Way: 1069011490

Version #2

Extend junction area to start of slip lane, as the slip lane only exists for the purpose of the junction

Edited about 1 year ago by [CycleStreets](#)
Changeset #156879733

Tags

area	yes
junction	yes

Nodes

► 38 nodes

[Download XML](#)

Way / [History](#) / [Versions](#):

1

East Road, Cambridge

www.openstreetmap.org/way/1069011490



An aerial photograph of a city street intersection. Overlaid on the map are two sets of lines representing street boundaries. One set consists of solid red lines with small white dots at the corners and white arrows indicating traffic flow. The other set consists of dashed white lines. The red lines form a more complex, slightly irregular shape, while the white lines form a more standard rectangular intersection. A white line with a bicycle symbol is also visible on the left side of the intersection.

Concept of a “Street” (two here)



Image: David Earl



Martin Lucas-Smith

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